

Directional Collapse in Fine-Tuning: Preserving the Way Back

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Abstract

Forward-only supervised fine-tuning can improve a trained mapping while damaging its backward counterpart. An apparent loss-based remedy may also change the directional information supplied during training, confounding objective and data effects. We test whether preservation requires a specialized loss or simply a small dose of backward examples across code transformation, format conversion, translation, semantic parsing, and algebra. Replacing just 1% of forward pairs with their reversals restores formatting backward success from 0.4% to 90.2%, while forward success remains 99.6% versus 99.7% in a fixed Llama3B comparison. Ordinary cross-entropy requires no new paired data and matches training instances and optimizer steps. Small-dose recovery also appears in code, translation, and semantic parsing, with task-dependent costs. Directional and extra-CE controls separate this baseline from contrastive and other auxiliary objectives; mixed-direction losses sometimes add value. Candidate-ranking and layer-intervention diagnostics support output-selection interference as an explanatory hypothesis, without establishing a universal mechanism. Adverse dose results, failed task gates, and a replicated unit-conversion null bound generality. Small-dose reversed-pair SFT is therefore a necessary baseline for attributing preservation to a new objective.

1 Introduction

Paired data carries a direction. A translation pair supports either language direction; a code transformation can be undone; and a question-SQL pair supports both parsing and verbalization. These tasks use the same pair but ask the model to generate different sides. Fine-tuning on one direction can improve that mapping while making the backward task fail, even when the base model already performs both. A forward-only evaluation would report success and miss the loss.

Prior work establishes this degradation in code transformation (Nikiema et al., 2025) and translation (Zhu et al., 2024). An auxiliary loss can appear to repair it while also introducing different paired examples or directional information. Matching total instances, tokens, or compute does not by itself hold that information fixed. We treat direction as an experimental factor: first ask what ordinary cross-entropy achieves with backward examples, then test what an auxiliary objective adds. The practical question is *how little backward training is needed to keep the backward task?* Every existing pair already supplies a target; replacing a small fraction with reversed pairs requires no new paired corpus.

The problem and the simple intervention. Table 1 puts base, forward-only SFT, and every registered adaptation side by side, with both directions expressed as success percentages. In its fixed Llama3B formatting comparison, SFT raises forward success from 60.2% to 99.7% but reduces backward success from 51.1% to 0.4%. Reversing 1% of training pairs yields 99.6% forward and 90.2% backward success. Code transformation shows the same qualitative pattern. These are concrete within-campaign examples, not a claim that every task improves forward or that 1% always suffices.

Why compare objectives against reversed pairs? A specialized loss can supply directional information as well as change optimization. Our contrasts separate those contributions using ordinary reversed-pair SFT, generic instruction replay, longer forward training, extra-CE proxies, and shuffled-pair controls. Table 2 shows both directions for contrastive and other auxiliary objectives. Forward-only alignment does not consistently rescue backward generation. Mixed-direction alignment sometimes beats the simple mixture, so our conclusion is that small-dose SFT is an essen-

Task	Dir.	Reference		Reversed-pair share					Other adaptations				
		Base	SFT	1%	5%	10%	25%	50%	Replay	Rev.	Fwd2×	Flip	Multi
German → English	F	73.7	67.1	67.0	67.4	66.2	68.6	67.4	68.6	63.5	57.8	64.4	69.1
	B	79.2	0.0	79.4	82.0	82.8	82.3	82.9	59.0	85.2	1.4	83.9	71.5
English → German	F	79.2	84.8	84.6	85.5	85.4	84.7	84.6	85.4	0.0	77.8	84.8	84.0
	B	73.3	64.4	70.8	68.5	68.2	68.2	67.8	72.0	66.2	64.4	67.0	73.0
English → Chinese	F	74.1	77.9	76.6	77.4	77.1	78.6	76.7	78.6	0.0	75.9	80.4	77.0
	B	75.4	4.0	68.2	68.3	69.9	71.5	71.9	70.9	71.2	9.5	72.8	70.3
Chinese → English	F	76.0	71.5	72.5	72.1	72.3	72.2	71.8	74.8	23.4	63.8	71.5	73.3
	B	74.0	0.0	76.2	77.1	80.0	78.3	77.2	1.0	77.7	0.0	76.1	71.0
Code transformation	F	37.5	82.4	82.8	84.3	83.6	84.4	83.9	74.8	65.6	88.3	86.3	70.5
	B	32.0	3.8	55.8	75.2	76.5	80.3	85.6	14.5	86.3	9.6	86.6	3.4
Question → SQL	F	49.5	62.4	62.3	60.8	62.0	60.1	58.6	59.8	42.0	61.1	60.8	56.8
	B	51.1	23.2	48.5	55.0	57.5	59.9	58.1	17.1	59.1	27.3	58.8	50.4
Format conversion	F	60.2	99.7	99.6	99.9	99.8	99.8	99.8	99.0	25.2	99.0	99.8	97.8
	B	51.1	0.4	90.2	96.2	97.0	98.6	99.6	24.6	99.4	0.7	99.6	42.9
Factual relations	F	71.9	86.7	–	–	–	86.7	87.4	88.1	79.3	89.6	89.6	83.0
	B	65.2	71.1	–	–	–	80.0	80.7	67.4	83.7	58.5	81.5	65.9
Expand → factor	F	17.3	26.1	25.5	26.8	27.3	25.5	24.8	25.8	14.6	29.6	27.7	21.2
	B	14.7	7.8	14.6	20.6	25.3	31.0	36.5	9.2	43.6	6.7	44.9	6.0
Factor → expand	F	14.6	43.6	44.0	43.8	43.2	40.2	36.1	36.9	7.8	49.6	44.7	26.8
	B	17.4	14.7	10.6	15.8	18.0	23.0	24.9	14.0	26.2	12.7	27.4	26.5
Execution prediction	F	26.5	29.0	–	–	–	29.5	29.5	27.0	21.0	31.5	30.5	24.5
	B	16.5	13.0	–	–	–	26.0	28.0	13.0	30.0	12.5	29.0	12.5

Table 1: Strict success (%); F/B denotes forward/backward. Blue highlights small reversed-pair doses; red highlights SFT backward success at most half the same-pass base (a descriptive threshold). Llama 3.2-3B, fixed seed 17, all eleven audited task orientations. Dose columns replace that share of forward pairs with their reversals, keeping instances and steps fixed. Replay replaces 50% with generic instructions; Rev. trains backward only; Fwd2× doubles forward training; Flip trains both directions at doubled budget; Multi mixes other tasks. Dashes are unresolved dose rungs, not zeros. The relation/execution low-dose extension (Amendment57) uses fewer than one effective batch of distinct reversed pairs; it is exploratory and excluded from the original registered knee analysis.

tial baseline, not that auxiliary objectives never help. Replacement matches instances and optimizer steps; realized token exposure and runtime are audited separately.

From a code-specific attribution problem to a broader preservation test. We ask three linked questions: does the problem and fix extend across task families; how small can the backward dose be; and what changes in the model’s behavior when the fix works? The main table tests code, serialization, translation, semantic parsing, and formal reasoning under the same directional intervention. The diagnostic evidence then examines answer competition and selective update interference. We distinguish these explanations from claims about memorization or permanent erasure, which backward failure alone cannot establish. Failed new-domain gates, replicated unit-conversion nulls, and recipe-dependent forward costs bound the claim. Model/seed coverage and paired uncertainty accompany the compact main tables in the appendix and source evidence.

2 Directional collapse, defined

A *paired task* is a pair of datasets (X, Y) and (Y, X) over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call $X \rightarrow Y$ the *forward* direction and $Y \rightarrow X$ the *backward* (or reverse) direction. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately:

Echo the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

Off-target the output is in the wrong language, format, or notation.

Reverse strict success excludes both failure modes. The original translation forward predicate combines target-language and quality checks but does not explicitly exclude echoes; its forward-retention interpretation has this limitation. The versioned transfer diagnostic in Amendment51 adds echo exclusion in both directions and retains the original failed gates. Serialization strict scoring excludes echoes in both directions.

Thresholds for continuous criteria are fixed from the *untouched* model’s own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

3 Setup

Domains. The audited main grid contains 11 task orientations covering translation, code transformation, semantic parsing, factual relations, executable reasoning, format conversion, and algebra. Opposite orientations of the same task are not independent domains. The invented-format never-had control is excluded from this count.

Models. The current audited results cover Llama 3B/8B, Gemma 4B/12B, and OLMo 1B. Eligibility and seed coverage vary by task and model; failed gates are retained. Appendix B.1 records split sizes, the reference training recipe, and realized training/evaluation counts.

Arms and budget. The forward-only baseline (SFT) trains on forward pairs. Each MIX k arm replaces $k\%$ of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count and optimizer steps are matched to SFT at every dose. We audit realized sequence and supervised-token counts separately: prompt length and unequal side lengths can change token exposure even when the pair count is fixed. Golovneva et al. (2024) distinguish data-matched from compute-matched reverse training; replacement requires no new paired examples. Any compute-saving claim is scoped to the measured training recipe and its audited token exposure.

Generic instruction REPLAY replaces 50% of forward pairs; its matched-dose comparison is with MIX50, while low-dose contrasts also differ

in replacement share. REV trains backward only and tests backward learnability, with its forward cost reported. Longer-forward and both-direction controls use additional training budget; all arm definitions and realized direction/token counts remain in the run manifests.

4 Forward gains can hide backward losses

Table 1 is the reference comparison: one fixed model and training seed, all eleven audited task orientations, and every registered main-grid adaptation. Paired forward/backward rows make the tradeoff visible without pooling tasks or choosing successful seeds. Red marks the registered descriptive collapse threshold; blue identifies low-dose mixtures, including their adverse results. Unresolved dose rungs are explicit.

The broader snapshot contains 91 distinct audited model–task–seed cells from 103 validated campaigns. Forward SFT leaves backward success at most half the same-pass base rate in 52 cells. Repeated passes are selected by latest valid completion time, independently of effect sign. These are descriptive counts, not independent-domain counts or pooled significance tests (Appendix A). Forward gains are not universal: translation orientations include forward degradation, while factual relations do not collapse in the reference comparison.

General-ability probes separately measure instruction following and reasoning, with overlap exclusions recorded before generation (Appendix B.3). Their endpoint-specific changes preclude interpreting task scores alone as disproportionate forgetting or universal preservation.

5 Small backward doses with ordinary SFT

The simplest preservation baseline changes only which side of an existing pair is the input. In the reference formatting cell, 1% reversal raises backward success from 0.4% to 90.2%; 5% reaches 96.2%. Forward success stays close to forward-only SFT. For code transformation, 1% raises backward success from 3.8% to 55.8% while forward success changes from 82.4% to 82.8%. Generic replay and doubled forward training do not provide comparable backward recovery in these same passes (Table 1). The contrast is particularly useful: more forward optimization is not

a substitute for correct backward targets.

Does it work outside code and formatting? In English-to-Chinese translation, 1% reversal raises backward success from 4.0% to 68.2%; forward success changes from 77.9% to 76.6%, exposing a retention cost. In question-to-SQL semantic parsing, backward success rises from 23.2% to 48.5% at 1%, and 55.0% at 5%; forward success at 1% is 62.3% versus 62.4% for SFT. For expansion-to-factorization algebra, 1% reversal raises backward success from 7.8% to 14.6%, close to the base’s 14.7%. Thus the pattern spans linguistic, executable, structured-output and symbolic tasks; the opposite algebra orientation retains its adverse low-dose result. These are fixed reference-cell comparisons, not evidence of equal effects across fields or training seeds.

The full dose ladder also shows why these examples cannot justify a universal 1% rule. Some tasks need higher doses; algebra includes an adverse low-dose result. We define the registered knee as the smallest dose reaching 90% of MIX50’s backward success, and report paired uncertainty and seed variation rather than treating adjacent rungs as equivalent. Figure 1 retains the units null alongside collapse cells. Forward retention, general-ability endpoints, measured runtime and token exposure are separate quantities; Appendix B.2 reports their joint audit.

6 Boundary conditions: information and inverse difficulty

The format ladder varies whether an inverse is determined, while algebra orientation and automata probe inverse search difficulty. Several ladder cells fail eligibility and cannot support tuned-model comparisons; their descriptive base results remain separate.

Two kinds of hardness. Information loss can prevent a unique inverse even with unlimited search, whereas a determined inverse can remain computationally difficult. Failed gates do not distinguish these explanations by themselves.

7 Why can backward examples help?

Different conditionals, identical underlying pairs. Forward CE trains the target y given forward instruction I_F and input x ; it supplies no direct penalty for an incorrect backward answer

given I_B and y . Let ℓ_θ be target-completion cross-entropy. Partition the original pairs into unreversed F_ρ and reversed B_ρ , where $|B_\rho|/|D|$ is the realized backward share. At the pair-distribution level, replacement has the schematic objective

$$\mathcal{L}_\rho = \frac{1}{|D|} \left[\sum_{(x,y) \in F_\rho} \ell_\theta(y | I_F, x) + \sum_{(x,y) \in B_\rho} \ell_\theta(x | I_B, y) \right]. \quad (1)$$

The relation between x and y is unchanged; the requested conditional changes. Our hypothesis is that backward examples maintain the appropriate task-conditioned output selection, counteracting a bias induced by forward-only updates. If the base already performs the inverse, a small dose may suffice to maintain that conditioning. This predicts preservation from ordinary CE without requiring a special alignment loss. It does not predict that every inverse is learnable or every model must collapse.

Evidence for output-selection interference. Five completed seed-17 diagnostics test this account (Appendix C). In formatting and translation collapse cells, forward SFT reduces preference for correct backward answers over copied inputs; mixed-direction SFT restores that preference. Gemma translation is particularly informative: absolute backward-answer loss improves relative to base despite collapsed generation. Correct-answer likelihood alone therefore fails to explain the behavior; relative output preference is a candidate explanation. Formatting middle-band removals recover backward success beyond scaling and random removals matched for retained update norm, with small forward changes. All bands and both directions remain reported, and edit norms are not also matched. This supports selective interference in one cell, not a universal causal account.

What the diagnostics rule out or leave open. Final-checkpoint gradient cosines provide little support for persistent gradient conflict. Local backward updates improve held-out loss even in the units null cell, so improvement alone is not a collapse-specific signature or a demonstrated generation repair. Total and per-token candidate rankings are both reported; candidate lengths and valid alternative translations limit their interpretation. Direction probes still need input-cue controls. Gemma diagnostics exclude inactive vision

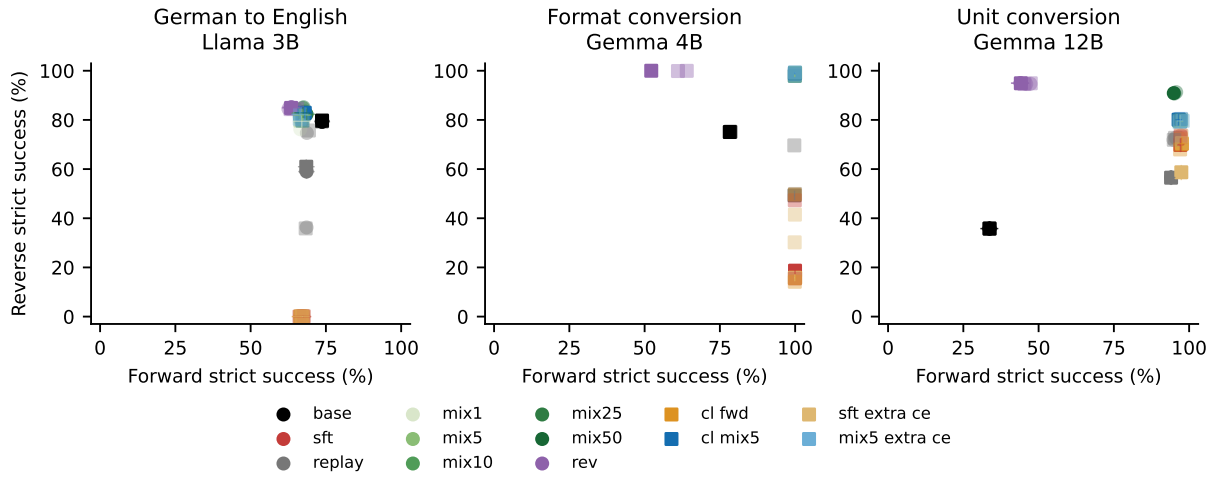


Figure 1: Current preservation frontiers on the fixed reporting panel. Circles show ordinary SFT/dose campaigns; squares show separately generated contrastive/control campaigns. Opaque points and marginal cluster-bootstrap intervals are seed-17 results; translucent points show other training seeds. Contrasts remain paired within each campaign and are retained with measured runtime/exposure records in the source-backed analysis. The units null-collapse boundary case is included. No cross-campaign pooled test is implied.

factors. Further registered diagnostics remain unmeasured and supply no evidence.

Recovery checks remain complementary.

Prompt elicitation, adapter scaling, and model/seed-matched relearning against a never-had control give separate verdicts. The never-had control uses a different invented-format task, and LoRA retains base weights by construction; neither behavioral recovery nor relearning alone proves stored knowledge. Spectral sweeps are exploratory. The separately registered full-weight projection repair has 1 saved checkpoint passing weight audits; its fresh campaign improves both directions. This is a scoped repair result, not a general remedy or an identified internal cause.

8 Consequences for attribution

We distinguish exposure to both paired sides from an objective-specific generation benefit.

A gain from an auxiliary objective can reflect its directional information, its additional computation, or the objective itself. Our paired contrastive arms combine forward-only or mixed-direction CE with symmetric representation alignment. Both sides are encoded independently, and negatives are drawn only from training data. The contrastive term has no reverse generation CE, but it still exposes inverse correspondence; it cannot be described as having no inverse supervision.

Table 2 reports fresh, complete passes for the registered contrastive comparison and aux-

iliary objectives that pass the current exposure/withdrawal audit. Each row pair has its own base; numerical differences across evaluation campaigns are not treatment effects. The contrastive grid currently completes 24 of 27 registered model-task-seed cells. Appendix A retains every registered primary contrast, per-seed uncertainty and exposure control.

Forward-only contrastive alignment provides no consistent benefit over extra-CE or shuffled controls. In the reference code cell, the reconstructed code-equivalence CFT recipe falls well short of 5% reversed-pair SFT on backward generation. CFT trains equivalence judgments with CE; it is distinct from the representation-alignment arms. Forward-only unlikelihood also fails to preserve backward translation in the audited cells. This supports trying backward examples before adding an auxiliary objective. It does not establish that every loss baseline is inferior: mixed-direction alignment sometimes beats 5% SFT, and round-trip training wins backward success in one SQL cell while trading forward performance. Extra-CE controls match an exposure proxy, not identical measured FLOPs. An interval spanning zero does not establish equivalence, and degenerate intervals at a generation floor require particular caution.

New-domain eligibility. The 6 original small-model task-model gates failed the unchanged eligibility criteria (Table 3). Complete test-set gates

(a) Contrastive objectives and exposure controls												
Task	Model	Dir.	Base	SFT	5%	Replay	Rev.	CL-F	CL-M	Shuffle	CE-F+	CE-M+
German → English	G12B	F	76.9	66.9	66.5	67.3	1.6	66.1	66.8	66.1	66.5	66.6
		B	72.0	0.2	60.1	56.2	66.4	0.2	60.9	0.3	0.1	60.4
German → English	G4B	F	75.4	67.7	67.4	68.7	40.0	67.2	69.4	67.1	66.9	66.7
		B	75.8	0.0	74.4	70.4	76.8	0.0	74.5	0.0	0.0	75.1
German → English	L8B	F	74.5	64.2	65.8	63.9	4.7	62.8	64.5	63.8	62.8	65.1
		B	73.0	16.5	68.9	64.3	75.1	0.0	67.7	0.0	18.2	69.6
German → English	L3B	F	73.7	67.3	67.0	68.5	63.4	66.5	68.0	66.7	67.0	66.4
		B	79.7	0.0	82.0	61.0	84.9	0.0	82.9	0.1	0.0	82.2
Format conversion	G4B	F	78.3	100.0	99.9	99.8	52.2	100.0	99.9	99.9	99.9	100.0
		B	75.1	18.7	98.0	49.3	100.0	15.8	99.0	45.0	15.8	99.1
Format conversion	L8B	F	80.2	99.5	99.9	99.8	2.2	100.0	99.8	99.7	100.0	99.9
		B	69.0	0.0	98.7	32.1	99.9	2.2	99.2	0.0	0.0	98.7
Format conversion	L3B	F	60.6	99.7	99.9	98.9	27.0	99.6	99.6	99.7	99.2	99.7
		B	51.4	0.4	96.0	25.6	99.4	0.1	96.0	0.8	0.6	94.2
Unit conversion	G12B	F	33.7	97.1	97.0	94.0	44.1	97.6	96.5	97.5	97.4	97.1
		B	35.8	69.8	79.4	56.5	94.9	70.3	80.1	74.1	58.7	79.6

(b) Other auxiliary objectives									
Task	Model	Dir.	Base	SFT	5%	CFT	Unlikelihood	Round-trip	
English → German	L8B	F	71.1	73.1	73.0	<i>n/a</i>	74.0	—	
		B	74.2	58.7	69.2	<i>n/a</i>	0.1	—	
English → German	L3B	F	79.7	84.8	85.4	<i>n/a</i>	85.1	—	
		B	72.8	63.6	68.8	<i>n/a</i>	0.0	—	
Code transformation	L3B	F	38.0	81.4	84.2	84.1	—	—	
		B	32.0	3.6	75.4	6.7	—	—	
Question → SQL	L8B	F	62.0	72.1	71.9	<i>n/a</i>	—	71.3	
		B	54.9	51.3	57.8	<i>n/a</i>	—	58.8	
Question → SQL	L3B	F	49.1	62.3	61.0	<i>n/a</i>	—	61.5	
		B	51.5	23.3	55.2	<i>n/a</i>	—	48.2	

Table 2: Strict success (%); F/B denotes forward/backward. Blue highlights small reversed-pair doses; red highlights SFT backward success at most half the same-pass base (a descriptive threshold). All eligible original contrastive and audited auxiliary-objective cells at fixed seed 17. Each task/model row pair uses its own fresh evaluation pass and base; scores across campaigns are not treatment contrasts. L/G denotes Llama/Gemma; B denotes billions of parameters. The 5% column is ordinary reversed-pair SFT. CL-F/M adds alignment to forward/mixed-direction CE; Shuffle randomizes positive pairs; CE-F+/M+ adds CE as an exposure proxy; CFT is the reconstructed code-equivalence contrastive fine-tuning recipe. Dashes denote pending compatible objectives; n/a marks code-equivalence CFT outside code. Failed gates, other seeds and paired intervals remain in the appendix/audit.

and scorer audits are retained. No tuned pilot is admitted for these failed cells. Explicit-output variants reuse their corpora; they are prompt diagnostics and cannot be counted as independent new-domain successes. The 6 completed explicit-format gates also fail the unchanged criteria. Larger-model feasibility gates on the original tasks are separately registered. Gemma12B passes the unit-conversion gate and completes valid replicated campaigns (Table 4). Forward-only SFT improves both directions; this is retained as a replicated null-collapse result. Replication seeds are admitted by validity, independently of effect sign. Llama8B also passes Python-C++ translation after a technical parser retry; its pilot is evaluated under the registered same-pass protocol. Boolean

gates remain failed, with correctness criteria unchanged. Completed versioned-contract pilots retain all seeds and controls (Appendix B.4). They include a typed Python/C++ Llama8B case where replay preserves backward behavior much better than the direction mixture, bounding the remedy beyond the main-grid examples.

Full-weight checks. Full-FT training is complete; 6 of 6 validated evaluation campaigns are complete in the current snapshot. Full checkpoints require isolated engines, unlike the single-engine multi-LoRA comparisons. Their newly generated campaigns use the same instances, messages, tokenizer, and scoring settings but span engine restarts. We record this layout and inter-

Task	Model	Forward	Reverse	Eligibility
units	llama32-3b	0.4	1.9	Fail
units	gemma3-4b	37.6	3.5	Fail
logic	llama32-3b	0.0	0.0	Fail
logic	gemma3-4b	0.0	0.0	Fail
py_cpp	llama32-3b	78.6	74.1	Fail
py_cpp	gemma3-4b	5.1	99.9	Fail

Table 3: Original new-domain base gates: strict success in percent. Failed cells are retained and receive no tuned pilot. Explicit-output diagnostics use the same semantic splits and are separate exploratory variants.

pret small differences against the decoding floor. Projection-matrix spectral repair is exploratory; it does not assume a validated noise model. Repair results require joint reverse/forward interpretation.

9 Related work

Prior directional degradation and reverse training are established starting points. Our contribution is the joint preservation comparison and controlled attribution, qualified by failed gates and replicated null-collapse results.

Directional degradation and reverse training. Nikiema et al. (2025) study cognitive specialization in code transformation; Zhu et al. (2024) examine direction-dependent degradation in translation. We ask how preservation varies across domains, doses, models, and objectives. Program-execution reasoning also exhibits directional asymmetry (Hasanov et al., 2026), motivating cell-specific measurement of starting reverse capability. The reversal curse concerns capabilities never acquired (Berglund et al., 2024); reverse training addresses this at the data level under data- and compute-matched comparisons (Golovneva et al., 2024). Our relearning experiment compares a collapsed model with a model-matched never-had control. Replacement mixtures hold instances and steps fixed while auditing realized exposure.

Forgetting and rehearsal. Broader forgetting after fine-tuning (Luo et al., 2023) and differences between LoRA and full fine-tuning (Biderman et al., 2024) motivate our general-ability and full-weight controls; they do not bound directional loss in these tasks. Rehearsal can preserve earlier performance (Scialom et al., 2022). Generic replay at the registered replacement share tests whether directional content adds preservation beyond that explanation. Concentration of inverse loss relative to general ability remains a separate empirical ques-

tion.

Recovery and task inference. Wrapper and offset accounts of fine-tuning (Jain et al., 2024; Lee et al., 2024), update filtering (spe, 2026), and skewed task inference (Kotha et al., 2024) motivate distinct recovery instruments. Our directional tests include a never-had control, both-direction retention, and fixed-threshold exploratory repair. Behavioral recovery does not identify a unique internal mechanism, and bidirectional behavior need not imply a unified representation (lat, 2026).

Attributing objective gains. Data mixtures and reporting choices can confound objective attribution (Longpre et al., 2023; Lipton and Steinhardt, 2018). RevThink adds backward questions and reasoning (Chen et al., 2025); its cross-entropy components illustrate why directional exposure must be controlled when assessing a bidirectional objective. Our paired contrastive comparison includes reversed-pair SFT, extra-CE proxies, and shuffled positives, measuring generation in both directions. These controls test the scope of an objective-specific gain without assuming that auxiliary objectives are universally unnecessary.

10 Conclusion

Forward-only SFT can improve a trained mapping while damaging an existing backward task. Changing data direction supplies a simple preservation baseline: small reversed-pair doses recover backward behavior across code, formatting, translation, semantic parsing, and selected algebra orientations using ordinary CE. The required dose and forward cost vary; nulls and auxiliary-objective gains remain visible. Candidate competition and selective layer interventions suggest output-selection interference as a hypothesis to test further. The practical rule is to evaluate both directions and hold directional information explicit before attributing preservation to a new

Task	Model	Seed	Arm	Forward	Reverse
py_cpp	llama31-8b	17	base	85.9	100.0
py_cpp	llama31-8b	17	sft	100.0	0.0
py_cpp	llama31-8b	17	mix50	100.0	48.7
py_cpp	llama31-8b	17	replay	100.0	97.6
py_cpp	llama31-8b	17	rev	0.0	0.0
py_cpp	llama31-8b	42	base	84.2	100.0
py_cpp	llama31-8b	42	sft	100.0	0.0
py_cpp	llama31-8b	42	mix50	1.7	0.0
py_cpp	llama31-8b	42	replay	99.9	99.5
py_cpp	llama31-8b	42	rev	0.0	0.0
units	gemma3-12b	1234	base	33.7	35.8
units	gemma3-12b	1234	sft	96.9	72.7
units	gemma3-12b	1234	mix50	95.1	90.7
units	gemma3-12b	1234	replay	94.8	71.9
units	gemma3-12b	1234	rev	47.3	94.7
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	mix50	94.9	90.9
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	rev	44.0	95.0
units	gemma3-12b	42	base	33.7	35.8
units	gemma3-12b	42	sft	97.3	73.7
units	gemma3-12b	42	mix50	95.7	91.3
units	gemma3-12b	42	replay	95.5	72.7
units	gemma3-12b	42	rev	45.9	94.6
py_cpp	llama31-8b	1234	base	84.2	100.0
py_cpp	llama31-8b	1234	sft	100.0	0.0
py_cpp	llama31-8b	1234	mix50	100.0	0.2
py_cpp	llama31-8b	1234	replay	95.8	97.9
py_cpp	llama31-8b	1234	rev	0.0	0.0

Table 4: Eligible larger-model generality campaigns: strict success in percent, with all arms and the base evaluated in one fresh pass per model/task/seed. The completed unit-conversion replications improve both directions under SFT and are retained null-collapse boundary cases. Additional eligible campaigns enter this table after validated analysis.

loss.

Limitations

Where the repair does not reach. Reversed-pair training cannot recover information absent from an input. Our failed new-domain gates also limit generality beyond established tasks.

Scale. The full arm set runs at 1–4B with three seeds; only core arms run at 8–12B with two where eligible in the original grid. Separately registered larger-model extensions add replication and objective controls in selected cells; coverage remains uneven. This does not establish scaling to larger models.

LoRA. Most arms use LoRA. Prior work reports different retention under full fine-tuning (Biderman et al., 2024); this does not guarantee a conservative directional-loss estimate in our tasks. The full-FT check covers only the registered Llama 3B seed-17 cells; it cannot bound directional loss across other models or training recipes.

Resolution of the dose ladder. Adjacent rungs differ by a couple of points, which our evaluation sets may not separate reliably. Dose comparisons use paired intervals and seed ranges; an interval spanning zero does not establish equivalence.

Criteria with a model in them. Two directions are scored by round-tripping through a frozen model rather than by execution. Each reports that model’s own accuracy on the references as the criterion’s ceiling where measured. Extractor/parser errors can reject valid generations; the D2T extractor fails its reference-text oracle, preventing a tuned comparison.

Languages and domains. The original code-transformation cell is Python only; the translation cells cover two language pairs, both with English on one side. Synthetic task variants do not establish natural code or unrestricted reasoning generality, and reused semantic corpora are not independent evidence. Independent transfer retains failed eligibility boundaries; completed prompt

Task	Seed	Direction	Base	Full SFT	Full mix5
code	17	forward	37.9	70.0	70.7
code	17	reverse	31.9	2.0	65.9
mt_de-en	17	forward	73.3	70.9	72.0
mt_de-en	17	reverse	80.1	72.8	83.6
mt_en-de	17	forward	80.1	87.0	86.6
mt_en-de	17	reverse	72.9	69.9	72.4
mt_en-zh	17	forward	73.9	79.1	79.2
mt_en-zh	17	reverse	75.9	66.9	71.8
mt_zh-en	17	forward	76.3	74.2	73.9
mt_zh-en	17	reverse	73.8	70.2	78.5
sql	17	forward	49.3	60.5	59.7
sql	17	reverse	50.4	18.7	55.1

Table 5: Completed Llama3B full-weight campaign rates in percent, with forward and reverse retention. Checkpoints use isolated engines and identical held-out instances; these comparisons span engine restarts. Paired intervals are recorded in followup contrasts. An empty body means no validated full campaign has completed.

and training-recipe tests retain adverse and null findings (Appendices B.4 and B.5).

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A Complete comparison counts and objective contrasts

The main tables show strict success percentages for a fixed reference seed. The following inventory covers the broader audited grid; contrastive tables retain all registered paired contrasts with their uncertainty and controls. Highlighting in the main tables identifies arms and the descriptive collapse threshold, not statistical significance.

B Additional evaluations, methods, and reporting audit

This appendix reports the exploratory additions registered before their measurements, their completed findings, and the boundaries that prevented

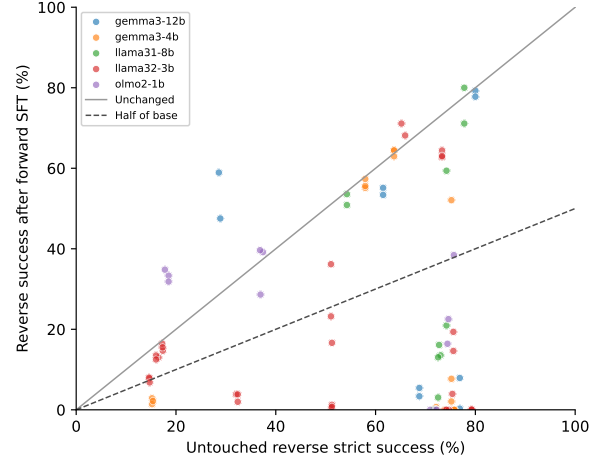


Figure 2: Audited model–task–seed cells, each using a same-pass base and SFT. The dashed line marks half the base rate. Repeated tasks and seeds are not independent observations.

eligible tuned comparisons. Allocation-local engineering smokes preceded production admission; passing a smoke alone did not establish a scientific result. The retained diagnostic subset comprises the five completed campaigns specified in Amendment54; broader registered diagnostics remain unmeasured. Failed gates and the replicated Gemma12B unit-conversion null-collapse result remain visible.

B.1 Paired data and reference training recipe

Each task supplies fixed training, validation, and test pairs. Content and pair-identity overlap checks exclude held-out test material during corpus construction; validation is not used to select a preferred test result. Opposite orientations of a task share the underlying task family. The reference recipe and actual corpus/trainer counts are reported in Table 10; smaller factual-relation and execution corpora are retained at their actual size. Seed-specific partitions define reversed-pair replacement. Training summaries retain realized supervised-token exposure and length filtering, rather than equating pair count with tokens. Every reported tuned comparison includes a freshly evaluated base and all declared arms on the same retained instance set.

Recorded reference recipe. The main-table Llama3B reference campaigns use the common recorded recipe: LoRA rank 32, scaling 64, dropout 0.05, learning rate 0.0001, 3 epochs, effective batch 64, and maximum sequence length 2048 tokens. The scheduler is cosine with warm-

Model	Task orientations	Cells	Threshold met
gemma3-12b	7	14	8
gemma3-4b	7	21	14
llama31-8b	6	12	7
llama32-3b	11	32	18
olmo2-1b	4	12	5

Table 6: Audited model–task–seed cells. Threshold met means SFT reverse strict success is at most half the same-pass base rate. Counts are descriptive and are not independent task counts or pooled significance tests.

Task	Model	Seed	Δ CL-fwd versus SFT	Δ CL-mix5 versus mix5
fnt	gemma3-4b	17	-2.88 [-3.88, -1.96]	+1.08 [+0.32, +1.88]
fnt	gemma3-4b	42	-1.28 [-1.96, -0.64]	-1.04 [-1.80, -0.32]
fnt	gemma3-4b	1234	+2.36 [+1.60, +3.20]	-0.32 [-0.92, +0.24]
fnt	llama31-8b	17	+2.24 [+1.52, +3.04]	+0.48 [-0.12, +1.12]
fnt	llama31-8b	42	-33.12 [-35.48, -30.64]	+0.40 [-0.20, +1.00]
fnt	llama31-8b	1234	+0.00 [+0.00, +0.00]	-0.72 [-1.52, +0.12]
fnt	llama32-3b	17	-0.24 [-0.56, +0.04]	+0.00 [-1.12, +1.12]
fnt	llama32-3b	42	-1.48 [-2.12, -0.92]	+1.68 [+0.52, +2.88]
fnt	llama32-3b	1234	-0.52 [-1.00, -0.04]	+4.44 [+3.08, +5.84]
mt_de-en	gemma3-12b	17	+0.00 [+0.00, +0.00]	+0.79 [-1.78, +3.56]
mt_de-en	gemma3-12b	42	-0.10 [-0.40, +0.00]	+0.59 [-2.08, +3.26]
mt_de-en	gemma3-12b	1234	+0.00 [+0.00, +0.00]	+0.69 [-1.68, +3.16]
mt_de-en	gemma3-4b	17	+0.00 [+0.00, +0.00]	+0.10 [-2.27, +2.27]
mt_de-en	gemma3-4b	42	+0.00 [+0.00, +0.00]	+0.40 [-2.08, +2.77]
mt_de-en	gemma3-4b	1234	+0.00 [+0.00, +0.00]	-0.30 [-2.47, +2.17]
mt_de-en	llama31-8b	17	-16.50 [-19.47, -13.64]	-1.19 [-4.05, +1.88]
mt_de-en	llama31-8b	42	-13.54 [-16.50, -11.07]	-0.40 [-3.26, +2.57]
mt_de-en	llama31-8b	1234	-23.32 [-26.78, -19.96]	-1.28 [-4.35, +1.58]
mt_de-en	llama32-3b	17	+0.00 [+0.00, +0.00]	+0.89 [-1.38, +3.26]
mt_de-en	llama32-3b	42	+0.00 [+0.00, +0.00]	+2.47 [+0.00, +5.04]
mt_de-en	llama32-3b	1234	-0.10 [-0.40, +0.00]	+0.10 [-2.57, +2.87]
units	gemma3-12b	17	+0.50 [-1.70, +2.80]	+0.70 [-0.90, +2.30]
units	gemma3-12b	42	-1.40 [-2.70, -0.20]	+0.40 [-1.10, +2.00]
units	gemma3-12b	1234	-4.90 [-7.00, -3.10]	+0.50 [-0.90, +1.90]

Table 7: Completed exploratory reverse-generation contrasts (percentage points), with 99% paired cluster-bootstrap intervals. Five primary contrasts are adjusted within each cell; no adjustment across cells. All proof-valid original and scale/domain-extension cells are included; failed-gate cells remain eligibility boundaries rather than tuned comparisons.

up fraction 0.03, weight decay 0 and gradient clipping 1. Training uses bfloat16 and final checkpoints without intermediate checkpoint selection. LoRA targets query, key, value and output attention projections and the gate, up and down MLP projections. Allocation-specific microbatches and accumulation retain the recorded effective batch. Full-weight and robustness variants are separate recipes, not substitutions for this reference.

B.2 Dose, forward retention, and computational cost

The main figure plots forward and reverse strict success for the base model, forward SFT, generic replay, available reversed-pair doses, and contrastive/control arms. Panels retain model, task orientation, and seed identity. Arms lacking a common evaluation campaign are not connected

as paired observations. The units boundary case is shown alongside collapse cells. Repeated passes use the latest validated completion time, with earlier passes retained in the audit; selection never uses effect sign.

Paired contrasts use the domain’s instance/cluster unit. Training-seed variation is separate: resampling test instances does not measure training variability. No aggregate pools cells as independent domains. The dose knee uses the criterion in §5; overlapping intervals do not establish equivalence.

The cost table reports available trainer runtime, optimizer steps, and CE-corpus tokens. Auxiliary encoded-side exposure is retained separately in the analysis. Missing timing, GPU-type, queue-time, or exposure records remain unavailable. Additional CE is the declared proxy; corpus token

Task	Model	Seed	CL-fwd vs extra CE	CL-fwd vs shuffled	CL-mix5 vs extra CE
fnt	Gemma 4B	17	+0.04 [-0.84, +0.96]	-29.20 [-31.80, -26.72]	-0.08 [-0.72, +0.56]
fnt	Gemma 4B	42	-16.04 [-18.00, -14.20]	-19.44 [-21.52, -17.48]	-0.44 [-1.28, +0.44]
fnt	Gemma 4B	1234	+8.20 [+6.80, +9.64]	-22.16 [-25.08, -19.24]	-0.60 [-1.24, +0.00]
fnt	llama31-8b	17	+2.24 [+1.52, +3.04]	+2.24 [+1.52, +3.04]	+0.44 [-0.16, +1.00]
fnt	llama31-8b	42	-21.52 [-23.64, -19.44]	-33.00 [-35.40, -30.52]	+0.16 [-0.32, +0.68]
fnt	llama31-8b	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.48 [-1.24, +0.28]
fnt	Llama 3B	17	-0.48 [-0.96, -0.08]	-0.68 [-1.16, -0.24]	+1.76 [+0.56, +2.96]
fnt	Llama 3B	42	-1.76 [-2.48, -1.12]	-3.40 [-4.32, -2.52]	+0.64 [-0.44, +1.68]
fnt	Llama 3B	1234	-0.36 [-0.88, +0.12]	+0.28 [+0.08, +0.60]	+2.76 [+1.48, +4.00]
mt_de-en	gemma3-12b	17	+0.10 [+0.00, +0.40]	-0.10 [-0.40, +0.00]	+0.49 [-2.17, +2.96]
mt_de-en	gemma3-12b	42	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	-1.19 [-3.56, +1.19]
mt_de-en	gemma3-12b	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.40 [-2.87, +2.08]
mt_de-en	Gemma 4B	17	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.59 [-2.77, +1.58]
mt_de-en	Gemma 4B	42	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [-2.27, +2.27]
mt_de-en	Gemma 4B	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+1.09 [-1.09, +3.36]
mt_de-en	llama31-8b	17	-18.18 [-21.34, -15.12]	+0.00 [+0.00, +0.00]	-1.88 [-5.04, +1.28]
mt_de-en	llama31-8b	42	-14.53 [-17.49, -11.96]	-0.20 [-0.59, +0.00]	-0.40 [-3.26, +2.47]
mt_de-en	llama31-8b	1234	-21.74 [-25.10, -18.38]	-0.10 [-0.49, +0.00]	-1.09 [-4.15, +1.68]
mt_de-en	Llama 3B	17	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+0.69 [-1.68, +3.16]
mt_de-en	Llama 3B	42	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+1.78 [-0.59, +4.25]
mt_de-en	Llama 3B	1234	-0.10 [-0.40, +0.00]	-0.10 [-0.40, +0.00]	-0.69 [-3.26, +1.88]
units	gemma3-12b	17	+11.60 [+8.50, +14.90]	-3.80 [-5.60, -2.00]	+0.50 [-1.00, +2.00]
units	gemma3-12b	42	-1.10 [-2.50, +0.20]	-0.10 [-1.80, +1.60]	+1.00 [-0.50, +2.60]
units	gemma3-12b	1234	-2.60 [-4.60, -0.80]	-4.80 [-7.00, -2.70]	+0.80 [-0.60, +2.30]

Table 8: The remaining three registered primary reverse-generation contrasts, in percentage points with 99% paired cluster-bootstrap intervals. Together with Table 7, these display all five primary comparisons for every completed cell. Extra CE is an exposure/compute proxy, not matched measured FLOPs. Degenerate zero intervals at an observed generation floor do not establish equivalence. Results remain exploratory; new scale/domain extensions are pending.

Registered evaluation	Model/task cells	Training seeds	Campaigns
General probes	3	17,42,1234	9
Prompt/template tests	4	17,42,1234	12
Independent transfer	2	17,42,1234	6
Recipe evaluations	2	17,42,1234	6

Table 9: Frozen paper-completion panel. Probe corpora retain 541 IFEval prompts and 1319 GSM8K questions; independent transfer retains 1688 OPUS translation pairs. Synthetic magnitude/template tests retain 1000 instances each. Recipe packs train 48 new adapters in isolated directories. Counts are planned workloads and audited corpus sizes, not completed results.

counts alone do not establish matched objective FLOPs or wall time.

Scoped finding. On the representative translation campaign, reversed-pair SFT preserves reverse behavior with nearly unchanged forward task performance relative to forward SFT (Table 11). Generic replay recovers less reverse performance in that campaign. The units case improves both ways under ordinary SFT; robustness results show that preservation and forward costs depend on task and recipe. Recorded budgets support within-recipe comparisons, not matched wall-time claims across different objectives.

B.3 General ability and overlap controls

The fixed panel is German-to-English/Llama3B, format conversion/Gemma4B, and units/Gemma12B. Each campaign compares base, SFT, replay, mixed-direction SFT, forward-only and mixed-direction contrastive arms, and their additional-CE controls. The auxiliary arm is fixed rather than selected by its best test score.

IFEval measures instruction following without demonstrations; GSM8K measures numerical reasoning using fixed training demonstrations. A neutral assistant system prompt is shared by all systems. Dataset revisions and overlap exclusions are recorded before generation. Auditing covers both sides of every local training/validation corpus, replay, and mixed-task sources, using normalized

Task orientation	Train	Val.	Test	Retained	Eval.	Steps	Batch $\times$ accum.
German $\rightarrow$ English	6500	500	1012	6500	1012	306	16×4
English $\rightarrow$ German	6500	500	1012	6500	1012	306	4×16
English $\rightarrow$ Chinese	6500	500	1012	6500	1012	306	4×16
Chinese $\rightarrow$ English	6500	500	1012	6500	1012	306	4×16
Code transformation	6500	500	2060	6499	2060	306	4×16
Question $\rightarrow$ SQL	6499	500	1034	6499	1034	306	16×4
Format conversion	6500	500	2500	6500	2500	306	16×4
Factual relations	504	69	135	504	135	24	16×4
Expand $\rightarrow$ factor	6500	500	2500	6500	2500	306	4×16
Factor $\rightarrow$ expand	6500	500	2500	6500	2500	306	4×16
Execution prediction	302	100	200	302	200	15	16×4

Table 10: Built paired-corpus sizes and recorded forward-SFT training/evaluation counts for the main-table reference campaigns. Train/validation/test counts are verified against the current frozen JSONL files; retained rows, steps and batch shapes come from each saved training manifest/summary. These are distinct provenance records: the code campaign retains fewer training rows than the current built corpus. Historical training data bytes are not inferred from a current file hash. Eval. counts paired instances per arm and direction. Opposite task orientations reuse task families; they are not independent datasets.

Task	Model	Seed	Arm	Forward (%)	Reverse (%)
mt_de-en	llama32-3b	17	base	73.7	79.2
mt_de-en	llama32-3b	17	sft	67.1	0.0
mt_de-en	llama32-3b	17	replay	68.6	59.0
mt_de-en	llama32-3b	17	mix5	67.4	82.0
mt_de-en	llama32-3b	17	mix50	67.4	82.9
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	mix50	94.9	90.9

Table 11: Representative seed-17 preservation rates. Each model/task row set comes from one complete campaign. Additional doses, seed variation, paired intervals and measured cost records are retained in the source-backed analysis. The units boundary case is included independently of effect sign.

exact/contained strings. Fixed demonstrations are checked against the retained test set. Lexical auditing does not establish semantic or pretraining independence; shared numerical structure limits interpretation of the units/GSM8K comparison.

All systems were evaluated together in fresh campaigns with adapter-effectiveness checks. Strict IFEval and strict GSM8K endpoints are reported separately; loose instruction checks and flexible numeric extraction remain secondary. Endpoint changes are paired within each campaign and seed variation is retained. A concentration claim is scoped to this declared panel; unrelated benchmark scales are not subtracted as if interchangeable.

Scoped finding. General retention depends on the endpoint and task (Table 13). Formatting/Gemma4B loses numerical reasoning under forward SFT while instruction-following changes are small; replay recovers more of that reasoning endpoint than mixed-direction SFT. Transla-

tion/Llama3B loses instruction following while improving numerical reasoning. Reverse-task recovery therefore does not establish general-ability preservation, and the evidence does not support a universal disproportionate-loss claim across benchmarks.

B.4 Independent real-data transfer

Existing German-to-English checkpoints are evaluated on the independently sourced OPUS-100 German-English test split.¹ The panel includes Llama3B and Gemma4B with the registered seeds and the same SFT, replay, mixed-direction, contrastive, and additional-CE systems. Dataset revision, character-length inclusion rules, directions, overlap exclusions, and scorer are frozen before generations. A new prompt on an existing semantic corpus remains a diagnostic rather than independent transfer evidence.

Overlap auditing includes all local training, val-

¹<https://huggingface.co/datasets/Helsinki-NLP/opus-100>.

Task	Model	Arm	Minutes	Steps	CE corpus tokens
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	mix50	9.0	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	cl_fwd	241.9	306	1096262
mt_de-en	llama32-3b	cl_mix5	62.2	306	1096262
mt_de-en	llama32-3b	sft_extra_ce	123.2	306	1096262
mt_de-en	llama32-3b	mix5_extra_ce	119.1	306	1096262
fnt	gemma3-4b	sft	30.6	306	1927990
fnt	gemma3-4b	mix5	30.7	306	1927882
fnt	gemma3-4b	replay	30.2	306	1927978
fnt	gemma3-4b	cl_fwd	194.3	306	1927990
fnt	gemma3-4b	cl_mix5	193.5	306	1927882
fnt	gemma3-4b	sft_extra_ce	191.3	306	1927990
fnt	gemma3-4b	mix5_extra_ce	194.0	306	1927882
units	gemma3-12b	sft	21.6	306	788819
units	gemma3-12b	mix50	22.1	306	788819
units	gemma3-12b	replay	21.5	306	799547
units	gemma3-12b	sft	21.6	306	788819
units	gemma3-12b	mix5	15.6	306	788819
units	gemma3-12b	replay	21.5	306	799547
units	gemma3-12b	cl_fwd	82.8	306	788819
units	gemma3-12b	cl_mix5	83.0	306	788819
units	gemma3-12b	sft_extra_ce	166.9	306	788819
units	gemma3-12b	mix5_extra_ce	168.0	306	788819

Table 12: Recorded seed-17 trainer runtimes (minutes), optimizer steps, and sequence tokens in the tokenized CE corpus (before epoch repetition). Auxiliary encoded-side exposure remains separately recorded in the analysis; these token counts are not total objective FLOPs. Runtime varies with GPU/training layout and is not a claim of matched wall time. Missing measurements are unavailable, never inferred.

idation, replay, and original test sources. Gold, echo, empty, and invalid-output oracles run on compute. Original model-specific COMET thresholds are retained. Strict success, continuous COMET, chrF++, echo, and off-target rates remain separate; checkpoint choice, decoding, and scoring are not tuned on this corpus.

The full fresh base was generated and scored first, with the resident engine reserved for tuned systems. Failure of the unchanged eligibility conjunction remains boundary evidence and blocks tuned generation. Small smokes establish engineering behavior, not production eligibility. Paired forward/reverse changes require an eligible campaign.

Completed measurements. Every registered production transfer campaign fails the unchanged echo-validity conjunction. The base-only boundary trials are retained in Table 14; no tuned checkpoint comparison is eligible. These measurements therefore do not establish preservation on this independent corpus, and the failed criterion is not evidence of adapter collapse.

Versioned diagnostics. Amendment51 registers an exploratory repair extension after inspection of failed cells. The original failed gates remain intact. Separate versions test bounded signed C++ literal syntax, explicit serialization/conversion/logic contracts, and a WebNLG extractor supplied with a global training-only predicate schema. Translation transfer additionally excludes source copies in both directions, retaining the original thresholds. Fresh compute-local oracles and base gates precede any tuned comparison. Revised criteria and prompts define new cells; their results do not replace original failures or support cross-version paired contrasts. Failed oracles and genuine capability limits remain reported boundaries.

Completed versioned-domain pilots. Table 15 reports every completed pilot under the revised contracts, including all registered seeds and both directions. Formatting/Llama8B and units/Gemma12B improve backward performance under forward-only SFT; their mixed-direction gains are additional backward learning rather than restoration of a collapsed capability. Typed

Cell	Endpoint	Seeds	SFT-base	mix5-SFT	Replay-SFT
fmt/Gemma4B	gsm8k	3	[-56.6, -36.3]	[+0.5, +21.8]	[+24.7, +46.0]
fmt/Gemma4B	ifeval	3	[-1.3, +0.6]	[-3.0, +0.9]	[-8.3, -4.1]
mt_de-en/Llama3B	gsm8k	3	[+6.6, +8.7]	[-2.7, +1.4]	[-13.7, +10.6]
mt_de-en/Llama3B	ifeval	3	[-10.9, -9.6]	[-0.4, +4.4]	[+0.2, +3.0]
units/Gemma12B	gsm8k	3	[+1.2, +2.0]	[-0.7, +0.9]	[-10.4, -9.1]
units/Gemma12B	ifeval	3	[-3.0, +1.5]	[+0.0, +3.5]	[-13.5, -3.0]

Table 13: Strict general-ability endpoint changes: descriptive minimum/maximum across all completed training seeds, in percentage points. Each difference uses its own fresh campaign baseline. These ranges are not confidence intervals. Per-seed paired intervals, all objective controls and overlap exclusions remain in the evidence artifact; endpoints are not pooled.

Model	Seed	Direction	Base (%)	Format failure (%)	Echo (%)	Campaign
gemma3-4b	1234	forward	33.9	10.1	6.6	blocked
gemma3-4b	1234	reverse	44.9	6.3	0.0	blocked
gemma3-4b	17	forward	33.9	10.0	6.6	blocked
gemma3-4b	17	reverse	44.8	6.3	0.0	blocked
gemma3-4b	42	forward	33.9	10.1	6.6	blocked
gemma3-4b	42	reverse	44.8	6.3	0.0	blocked
llama32-3b	1234	forward	33.7	10.6	6.8	blocked
llama32-3b	1234	reverse	53.0	6.8	0.0	blocked
llama32-3b	42	forward	33.7	10.6	6.8	blocked
llama32-3b	42	reverse	52.9	6.8	0.0	blocked
llama32-3b	17	forward	33.7	10.6	6.8	blocked
llama32-3b	17	reverse	52.8	6.8	0.0	blocked

Table 14: Completed OPUS transfer eligibility measurements with unchanged thresholds. Each campaign requires both directions to satisfy the frozen rate, format-failure, and echo conjunction. Failed campaigns preserve base-only trials and do not generate tuned comparisons; an echo failure is a criterion-validity boundary, not evidence of adapter collapse.

Python/C++ at Gemma12B recovers backward success with mixed-direction SFT, with seed-dependent forward costs. At Llama8B, the mixture fails to recover backward behavior consistently, whereas generic replay retains it much better; backward-only training also fails. These results bound the remedy and separate backward learnability from directional preservation. This panel tests only the registered half-reversal mixture, so it supplies no evidence for a small-dose optimum. Revised boolean gates remain failed. The WebNLG extractor fails its reference-text oracle, so no D2T tuned comparison is eligible.

B.5 Prompt and training-recipe robustness

The prompt panel includes translation/Llama3B, format conversion/Gemma4B, units/Gemma12B, and Python-C++/Llama8B. Frozen variants retain the original primary instruction and add fixed task/contract prefixes. Both directions use the same semantic instances, decoding limits, and scorers. Every frozen prompt is retained, separating echo/off-target output, semantic error, and measurement failure. Best-prompt summaries re-

main exploratory recovery diagnostics.

Synthetic extrapolation additionally uses exact unit magnitudes outside original support and behavior-disjoint piecewise-quadratic Python/C++ functions. Original allowlisted scorers and input ranges remain unchanged; gold outputs are audited on all generated instances on compute. These tests do not establish unrestricted natural-code, loop/list, or independent-domain generality.

The recipe panel fixes lower/higher learning-rate and LoRA-rank variants around the original recipe, varying the factors separately. Parameter values, cells, seeds, and primary contrasts are fixed in the recorded protocol. The comparisons preserve alpha/rank ratio, effective batch, paired-data identity, direction mixture, sequence limit, epoch count, and optimizer schedule. The audit records realized exposure, step count, and available GPU cost. SFT and mixed-direction SFT are compared on format conversion/Llama3B and units/Gemma12B, retaining collapse and null cases.

New adapters were stored separately. The original arms, base, and all recipe variants were regen-

Versioned task	Model	Seed	Dir.	Base	SFT	50%	Replay	Rev.
Formatting contract	L8B	17	F	34.4	99.9	99.9	99.9	0.0
			B	60.0	63.0	100.0	54.8	99.9
Formatting contract	L8B	42	F	34.9	99.9	99.9	99.7	0.1
			B	60.9	61.6	100.0	16.5	99.9
Formatting contract	L8B	1234	F	34.4	99.9	100.0	99.7	1.1
			B	60.0	66.6	100.0	15.7	99.9
Units contract	G12B	17	F	39.9	97.0	95.1	92.2	38.7
			B	46.9	70.3	89.9	65.0	95.7
Units contract	G12B	42	F	39.9	97.3	93.8	92.9	27.2
			B	46.9	73.3	88.0	73.0	95.6
Units contract	G12B	1234	F	39.9	97.2	94.4	94.4	9.1
			B	46.9	71.2	88.6	70.4	95.9
Typed Python/C++	G12B	17	F	86.2	100.0	100.0	9.2	0.0
			B	95.0	35.0	100.0	0.0	100.0
Typed Python/C++	G12B	42	F	86.2	100.0	85.0	36.1	0.0
			B	95.0	72.7	100.0	30.5	100.0
Typed Python/C++	G12B	1234	F	86.2	74.8	73.5	2.3	0.0
			B	95.0	55.8	100.0	0.0	100.0
Typed Python/C++	L8B	17	F	84.2	100.0	100.0	99.6	0.0
			B	100.0	0.0	48.4	99.4	0.0
Typed Python/C++	L8B	42	F	85.9	100.0	1.8	99.9	0.0
			B	100.0	0.0	0.0	99.4	0.0
Typed Python/C++	L8B	1234	F	85.9	100.0	100.0	95.9	0.0
			B	100.0	0.0	0.1	97.8	0.0

Table 15: All completed Amendment51 versioned-domain pilot campaigns, strict success (%). Each F/B pair uses its own fresh same-pass base and all registered controls; seeds are reported separately. These post-inspection contracts reuse the original semantic corpora and are exploratory diagnostics, not independent task replication. Only the 50% reversal dose was tested here. Red uses the descriptive SFT backward-loss threshold from the main table. Original failures remain reported separately; no cross-version subtraction or best-seed selection is used.

erated together in each campaign. Tables retain every adverse/null outcome, paired intervals and training-seed ranges; no best-recipe selection supports the primary claim.

Completed measurements. The frozen prompt panel retains directional loss under forward SFT for translation and format conversion; unit conversion remains a null-collapse case under the original recipe (Table 16). Preservation is not uniform across tasks or recipes: the Python-C++ mixture has large seed-dependent forward costs, and the higher learning-rate unit recipe includes reverse loss as well as improvement (Table 17). The unit boundary case is therefore scoped to the original recipe. Every frozen prompt and recipe is retained; selecting the best variant does not support the primary claim. These seed ranges are descriptive, and endpoint scales are not pooled into an overall preservation score.

B.6 Contribution and reproducibility

The abstract and conclusion state the supported contribution relative to prior directional degrada-

tion and reverse-training work: the preservation frontier, direction and exposure controls for objective attribution, and limits revealed by gates and null cells. Generality, disproportionate loss, inexpensive preservation, and recovery are separate claims with separate evidence requirements. A favorable contrastive result supports a scoped objective contribution; a null or generation-floor result does not establish equivalence.

Supported contribution. The study tests preservation of an existing reverse capability using ordinary supervised training with direction and exposure controls. Its contribution is the audited comparison of both task directions, objective attribution against matched baselines, and explicit limits from failed gates, null cells and sensitivity to training recipes. Exploratory recovery diagnostics qualify hypotheses about accessibility; they do not establish a universal internal mechanism.

The local review package includes recorded corpora and versioned reconstruction metadata; overlap and scorer audits; manifests and environment settings; irreplaceable per-instance gen-

Cell	Setting	Seeds	ΔF_{S-B}	ΔR_{S-B}	ΔF_{M-S}	ΔR_{M-S}
fmt/G4B	original/contract_first	3	[+20.0, +20.2]	[-58.5, -27.8]	[+0.0, +0.1]	[+51.9, +82.1]
fmt/G4B	original/primary	3	[+21.4, +21.7]	[-60.0, -27.9]	[+0.0, +0.2]	[+51.8, +83.8]
fmt/G4B	original/reword	3	[+22.2, +22.2]	[-59.8, -27.8]	[-0.0, +0.1]	[+51.8, +83.5]
mt_de-en/L3B	original/contract_first	3	[-4.4, -4.2]	[-79.0, -78.7]	[-0.5, +0.6]	[+79.9, +81.1]
mt_de-en/L3B	original/primary	3	[-7.7, -5.5]	[-79.7, -79.6]	[-0.3, +0.7]	[+79.2, +82.0]
mt_de-en/L3B	original/reword	3	[-4.6, -3.1]	[-79.1, -78.9]	[-0.7, +0.5]	[+78.8, +81.6]
py_cpp/L8B	original/contract_first	3	[+4.4, +4.4]	[-100.0, -100.0]	[-92.5, +0.0]	[+0.0, +50.0]
py_cpp/L8B	original/primary	3	[+15.8, +15.8]	[-100.0, -100.0]	[-97.8, +0.0]	[+0.0, +48.7]
py_cpp/L8B	original/reword	3	[+6.4, +6.4]	[-100.0, -100.0]	[-89.2, +0.0]	[+0.0, +49.9]
py_cpp/L8B	py_cpp_pieewise_quadratic/primary	3	[-68.2, -68.2]	[+0.0, +0.0]	[+0.0, +0.0]	[+99.3, +100.0]
units/G12B	original/contract_first	3	[+63.6, +63.6]	[+32.5, +36.6]	[-2.3, -1.4]	[+17.1, +20.7]
units/G12B	original/primary	3	[+63.3, +63.3]	[+34.0, +38.0]	[-2.1, -1.4]	[+17.4, +21.3]
units/G12B	original/reword	3	[+65.3, +65.6]	[+47.3, +50.7]	[-2.3, -1.6]	[+17.4, +20.6]
units/G12B	units_magnitude/primary	3	[+35.1, +47.2]	[+36.7, +40.0]	[-1.4, +5.0]	[+6.0, +8.5]

Table 16: Completed robust panel: ranges across training seeds of within-campaign strict-success changes (percentage points). S , B , and M denote forward SFT, same-pass base, and the registered directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

Cell	Setting	Seeds	ΔF_{S-B}	ΔR_{S-B}	ΔF_{M-S}	ΔR_{M-S}
fmt/L3B	lr_high	3	[+39.5, +39.5]	[-51.1, -49.2]	[-0.2, -0.0]	[+95.4, +98.2]
fmt/L3B	lr_low	3	[+38.7, +39.1]	[-43.0, -40.6]	[+0.2, +0.6]	[+82.6, +86.6]
fmt/L3B	original	3	[+39.4, +39.5]	[-50.8, -49.8]	[-0.1, +0.2]	[+92.2, +95.7]
fmt/L3B	rank_high	3	[+39.0, +39.3]	[-51.1, -51.0]	[-0.0, +0.4]	[+95.8, +96.6]
fmt/L3B	rank_low	3	[+38.2, +39.4]	[-50.4, -45.4]	[-0.0, +1.1]	[+90.8, +92.9]
units/G12B	lr_high	3	[+63.6, +63.7]	[-28.2, +12.1]	[-4.2, -1.1]	[+43.7, +84.6]
units/G12B	lr_low	3	[+62.8, +63.1]	[+36.1, +37.4]	[-3.4, -2.5]	[+14.1, +16.2]
units/G12B	original	3	[+63.3, +63.6]	[+34.1, +37.8]	[-2.3, -1.2]	[+17.7, +21.2]
units/G12B	rank_high	3	[+63.4, +64.3]	[+13.7, +27.9]	[-2.7, -1.9]	[+27.7, +41.1]
units/G12B	rank_low	3	[+62.8, +63.6]	[+25.6, +37.6]	[-3.1, -0.4]	[+15.9, +28.3]

Table 17: Completed recipe panel: ranges across training seeds of within-campaign strict-success changes (percentage points). S , B , and M denote forward SFT, same-pass base, and the registered directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

erations and their hashes; analysis entrypoints; and figure/table provenance. Failed gates, withdrawals, and corrected replacements remain traceable. Dataset/weight redistribution follows their licenses.

The source-backed generators revalidate trial hashes, complete paired coverage, successful evaluation statuses, and adapter effectiveness before rendering the result tables. The local bundle records original and packaged hashes after path redaction and supplies independent hash verification and strict-mean replay. Its manuscript folder builds separately from the experiment workspace and passes the strict review-format checks. These checks establish local result and document reproducibility, not reconstruction of historical training. Missing historical code/data versions and

upstream dataset license/reconstruction requirements remain explicit external-release requirements. Model and adapter weights are excluded. No public artifact release or conference submission has been performed.

C Budget-limited explanatory diagnostics

Amendment54 limits new compute to five fixed seed-17 diagnostics and required engineering smokes. The original 24-job panel remains registered; unrun jobs are deferred. Selection uses the time budget and distinct scientific questions, not new diagnostic outcomes. Completed earlier seeds are retained. These single-seed analyses are exploratory; paired intervals measure test-pair uncertainty, not training-seed stability. All candidates,

Deliverable	Required evidence	Current status
Preservation figure	Both directions, intervals, seeds, doses, exposure and measured cost	Audited synthesis and scoped interpretation integrated
General-ability table	Frozen panel, overlap exclusions, same-pass changes and intervals	All frozen campaigns complete; endpoint-specific findings integrated
Independent transfer	Corpus/version, eligibility, both directions, failures and hashes	All frozen OPUS gates fail; base-only boundaries retained
Robustness table	Frozen prompts/recipes, both directions, costs and all outcomes	All prompt/recipe campaigns complete; scoped results integrated
Reproducibility manifest	Splits/scorers, environment, manifests, trial hashes and analysis commands	Local hash/replay and standalone-build checks; external release requirements recorded

Table 18: Evidence coverage audit. Registration, auditing, and queue admission are separate from completed experimental evidence.

controls, bands and local steps remain in the evidence bundle. Proof-valid diagnostic jobs at this snapshot: 5 of 5. In the completed Llama formatting and translation diagnostics, forward SFT lowers gold completion loss in the forward direction and raises it in reverse; mixed-direction SFT lowers reverse loss relative to SFT with small mean forward changes (Table 19). The local SFT gradient cosines are near zero (Table 20); these checkpoints provide little support for a simple persistent gradient-conflict account. Gradients at these final checkpoints do not reconstruct interference along the training trajectory. This descriptive evidence concerns answer likelihood and does not establish stored knowledge. The Gemma translation diagnostic differs: reverse gold completion loss improves relative to base despite collapsed free generation. Absolute answer loss therefore does not uniformly track collapse. In the three collapse cells, per-token gold-over-copy preference falls under forward SFT and recovers under mixed-direction SFT; the units null cell retains strong preference (Table 21). Total and per-token rankings are reported together because candidate lengths differ. Translation has multiple valid answers; a restricted candidate ranking is neither correctness nor a measure of retained knowledge. At the SFT checkpoint, both tested reverse and mixed local-step sizes lower reverse held-out loss in every cell, including the null cell; this is not a collapse-specific signature or a demonstrated generation repair. All checkpoint states, update directions, step sizes and both endpoints are retained in the bundle. Formatting middle-band removals im-

prove reverse success beyond all matched controls with small forward changes, whereas the earliest band harms forward performance without consistent reverse recovery (Table 22). This supports depth selectivity in one cell, with unadjusted intervals across multiple interventions, rather than a universal causal mechanism.

1004
1005
1006
1007
1008
1009
1010

Task	Model	Direction	SFT minus base	mix5 minus SFT
fmt	llama32-3b	forward	-0.117 [-0.132, -0.101]	+0.000 [-0.000, +0.001]
fmt	llama32-3b	reverse	+0.121 [+0.104, +0.139]	-0.288 [-0.331, -0.247]
mt_de-en	llama32-3b	forward	-0.277 [-0.336, -0.220]	+0.004 [-0.003, +0.012]
mt_de-en	llama32-3b	reverse	+0.103 [+0.075, +0.131]	-0.181 [-0.206, -0.155]
mt_de-en	gemma3-4b	forward	-2.352 [-2.547, -2.168]	-0.002 [-0.012, +0.008]
mt_de-en	gemma3-4b	reverse	-1.008 [-1.122, -0.893]	-0.419 [-0.459, -0.378]
units	gemma3-12b	forward	-2.587 [-2.829, -2.358]	+0.016 [-0.005, +0.053]
units	gemma3-12b	reverse	-2.612 [-2.832, -2.405]	-0.381 [-0.503, -0.273]

Table 19: Gold completion NLL per token, paired differences with descriptive 95% bootstrap intervals on 128 fixed pairs. Negative values favor the first system. Candidate likelihood does not establish generation correctness or stored knowledge. All seeds here are 17.

Task	Model	SFT	Replay	mix5
fmt	llama32-3b	-0.018	+0.004	+0.010
mt_de-en	llama32-3b	-0.000	+0.048	+0.059
mt_de-en	gemma3-4b	+0.021	+0.012	+0.031
units	gemma3-12b	-0.013	+0.009	-0.026

Table 20: Forward/reverse validation-gradient cosine on 16 pairs in local LoRA A/B coordinates. Gemma excludes inactive vision factors. These coordinate-dependent quantities do not by themselves establish interference or causality.

Task	Model	Direction	Base	SFT	mix5
fmt	llama32-3b	forward	0.695 / 0.719	1.000 / 1.000	1.000 / 1.000
fmt	llama32-3b	reverse	0.750 / 0.672	0.000 / 0.008	1.000 / 1.000
mt_de-en	llama32-3b	forward	0.344 / 0.195	0.484 / 0.234	0.445 / 0.242
mt_de-en	llama32-3b	reverse	0.062 / 0.180	0.000 / 0.000	0.320 / 0.562
mt_de-en	gemma3-4b	forward	0.375 / 0.289	0.461 / 0.398	0.359 / 0.273
mt_de-en	gemma3-4b	reverse	0.406 / 0.586	0.000 / 0.000	0.430 / 0.562
units	gemma3-12b	forward	0.828 / 0.875	1.000 / 1.000	0.992 / 1.000
units	gemma3-12b	reverse	0.586 / 0.547	1.000 / 0.977	1.000 / 1.000

Table 21: Fraction of 128 fixed pairs where gold completion NLL is strictly lower than copied-input NLL: total / per-token ranking. Candidate lengths differ, so these rankings need not agree. Both directions and the same focal systems as the likelihood contrasts are shown; all eight systems and shuffled-target candidates remain in the bundle. Single training seed 17.

Removed band	Matched control	Direction	Success difference
band0_remove	band0_scale	forward	-0.043 [-0.062, -0.025]
band0_remove	band0_scale	reverse	-0.006 [-0.014, +0.000]
band0_remove	band0_random101	forward	-0.039 [-0.057, -0.021]
band0_remove	band0_random101	reverse	-0.014 [-0.023, -0.004]
band0_remove	band0_random202	forward	-0.037 [-0.055, -0.020]
band0_remove	band0_random202	reverse	+0.000 [+0.000, +0.000]
band1_remove	band1_scale	forward	-0.010 [-0.021, +0.000]
band1_remove	band1_scale	reverse	+0.096 [+0.070, +0.121]
band1_remove	band1_random101	forward	-0.008 [-0.018, +0.000]
band1_remove	band1_random101	reverse	+0.094 [+0.068, +0.119]
band1_remove	band1_random202	forward	-0.008 [-0.018, +0.000]
band1_remove	band1_random202	reverse	+0.100 [+0.074, +0.125]
band2_remove	band2_scale	forward	-0.002 [-0.012, +0.006]
band2_remove	band2_scale	reverse	+0.084 [+0.061, +0.107]
band2_remove	band2_random101	forward	+0.000 [-0.010, +0.010]
band2_remove	band2_random101	reverse	+0.082 [+0.057, +0.107]
band2_remove	band2_random202	forward	+0.002 [-0.008, +0.012]
band2_remove	band2_random202	reverse	+0.088 [+0.062, +0.113]
band3_remove	band3_scale	forward	+0.002 [+0.000, +0.006]
band3_remove	band3_scale	reverse	+0.037 [+0.021, +0.055]
band3_remove	band3_random101	forward	+0.006 [+0.000, +0.014]
band3_remove	band3_random101	reverse	+0.033 [+0.018, +0.051]
band3_remove	band3_random202	forward	+0.008 [+0.002, +0.016]
band3_remove	band3_random202	reverse	+0.043 [+0.025, +0.061]

Table 22: All fixed-depth removals versus retained-norm-matched controls for formatting/Llama3B (seed17), strict no-echo success differences on 512 pairs with descriptive 95% paired intervals. Retained and edit norms are both recorded; only retained norms are matched. No best band or control is selected.